Group II Row IV (full cross section): DGLAP, running coupling and finite terms, step by step

1 Result

Your three lines are (L1), (L2), (L3) of Eq. (3.1). Exactly, up to $O(\Lambda/k^+)$:

$$\begin{aligned}(\mathrm{L1})+(\mathrm{L2})+(\mathrm{L3}) = {}& \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\,\mathrm{LO}\big[\mathcal{B}\big]}_{\text{JIMWLK list}} + \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{\vee}{k^+}\,\mathrm{LO}\big[\mathcal{B}\big] + \log\frac{\vee}{k^+}\,H_4}_{\text{BFKL list}} \\ & + \underbrace{\Big(\text{pole}\to D_{h/g}\Big) - \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}(\xi)\,\mathrm{LO}_\xi\Big[\log\frac{(y'-y)^2\mu_F^2}{c_0^2}\Big] - 2N_c\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{\mathrm{sub}}\big[\log(y'-y)^2\big]\Big\}}_{\text{DGLAP}} \\ & + \underbrace{\frac{g^2\beta_0}{16\pi^2}\,\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big]}_{\text{running coupling (after the coupling counterterm)}} \underbrace{-\Big(1-\frac{k^+}{\vee}\Big)H_4 + \frac{g^2N_c}{4\pi^2}\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{\mathrm{lp}}\Big|_{d_\perp=2}}_{\text{finite}}. \end{aligned} \tag{1.1}$$

Notation is in Sec. 2: $a=\frac{k^+}{\vee}$, $\mathcal{B}=-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)$. In the DGLAP term the $\mathrm{LO}_\xi^{\mathrm{sub}}$ term equals $-\big[\log\frac{\vee}{k^+}-1+\frac{k^+}{\vee}\big]H_4$, Eq. (4.9).
Three remarks:

- The two evolution terms carry the ϵ-bracket $\mathcal{B}$. It is the $d_\perp$-dimensional transverse integral of the kernels of this row. Its pole cancels between the rows that make up JIMWLK⊗LO and BFKL⊗LO (`full_check`); it is not DGLAP and not running coupling.
- The running coupling is $\mathrm{LO}|_{\alpha_s(\mu_R)}+\mathrm{RC}=\mathrm{LO}|_{\alpha_s\to\alpha_s(c_0/|y'-y|)}$, inside the integral.
- At leading power ($a\to 0$): $\int_a^1\to\int_0^1$ and $(1-a)H_4\to H_4$.

All steps are checked in `checks.py` (Sec. 5).

2 Notation

$d_\perp=2-\epsilon$; $c_0=2e^{-\gamma_E}$; $a\equiv\frac{k^+}{\vee}$; $\xi_\Lambda\equiv\frac{k^+}{k^++\Lambda}$; $\beta_0=\frac{11}{3}N_c$; $\frac{g^2N_c}{4\pi^2}=\frac{\alpha_s}{2\pi}2N_c$; $\frac{g^2}{8\pi^2}=\frac{\alpha_s}{2\pi}$.

$$\ell_\Delta(\mu)\equiv 2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}=\log\frac{(y'-y)^2\mu^2}{c_0^2}, \tag{2.1}$$

$$\begin{aligned}\mathrm{LO}_\xi[f]\equiv{}& \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,f(y'-y)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} \\ & \times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{2.2}$$

$\mathrm{LO}\equiv\mathrm{LO}_{\xi=1}[1]$. The weight f is *inside* the integral.

- $\mathrm{LO}_\xi^{\mathrm{lp}}$: Eq. (2.2) with $f=1$ and the kernel of your (L3).
- $\mathrm{LO}_\xi^{\mathrm{sub}}[f]$: Eq. (2.2) with $\frac{(x'-y')}{(x'-y')^2}\to\frac{(x'-y)}{(x'-y)^2}$ and $U^{e'c}(y')\to U^{e'c}(y)$, i.e. the point y' is kept only in the phase and in f.

$$\mathrm{LO}^y\equiv\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}\big[U^{e'c}(y)-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\qquad H_4=\frac{g^2N_c}{\pi\,\mathbf{k}^2}\,\mathrm{LO}^y, \tag{2.3}$$

H_4 being your term $\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5k^+}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}[\dots]$ $(y\leftrightarrow z)$: the w,w' integrals give $\frac{4\pi^2}{\mathbf{k}^2}$.

$$\mathcal{P}_{gg}(\xi)=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{\beta_0}{2}\delta(1-\xi),\qquad \int_a^1 d\xi\,\frac{\xi\,G(\xi)}{(1-\xi)_+}\Big|_{\mathrm{std}}\equiv\int_a^1 d\xi\frac{\xi G(\xi)-G(1)}{1-\xi}+G(1)\log(1-a). \tag{2.4}$$

3 The starting point

Your expression, in this notation (the ϵ-bracket is a function of $y'-y$, so it is a weight inside the integrals):

$$\begin{aligned}
\text{(L1)} &= \frac{g^2N_c}{4\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]\mathrm{LO}_\xi\big[\mathcal{B}_0\big],\\
\text{(L2)} &= \frac{g^2N_c}{4\pi^2}\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\mathrm{LO}\big[\mathcal{B}_0\big],\\
\text{(L3)} &= \frac{g^2N_c}{4\pi^2}\int_a^{\xi_\Lambda} d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}^{\mathrm{lp}}_\xi,
\end{aligned}\tag{3.1}$$

with $\mathcal{B}_0 \equiv -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_{\overline{MS}}^2}{4} - 1$.

Your plus prescription in (L1). It is $\int_a^1 \frac{d\xi\,\xi G}{(1-\xi)_+} = \int_a^1 \frac{\xi G(\xi)-G(1)}{1-\xi}$, with no $\log(1-a)$ (with $G = \mathrm{LO}_\xi[\mathcal{B}_0]/\xi^2$, so $G(1)=\mathrm{LO}[\mathcal{B}_0]$). This is what the cut $\xi\le\xi_\Lambda$ ($p^+\ge\Lambda$) gives:

$$\int_a^{\xi_\Lambda} d\xi\,\frac{\xi\,G(\xi)}{1-\xi} = \int_a^1 d\xi\,\frac{\xi G(\xi)-G(1)}{1-\xi} + G(1)\Big[\log\frac{k^+}{\Lambda}+\log(1-a)\Big] + O\Big(\frac{\Lambda}{k^+}\Big).\tag{3.2}$$

These are the $\log\frac{k^+}{\Lambda}$ and the $\log(1-a)$ of (L2). (Checked: difference 2.6×10^{-5} at $\Lambda/k^+ = 10^{-5}$.)

4 Step by step

Step 1: the ϵ-bracket. With $d_\perp = 2-\epsilon$, $-\frac{d_\perp}{\epsilon} = -\frac{2}{\epsilon}+1$, so the -1 cancels and, exactly,

$$\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\Big[2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\Big] = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta(\mu) \equiv \mathcal{B}.\tag{4.1}$$

Step 2: split (L2). $\log\big(\frac{\vee}{k^+}-1\big) = \log\frac{\vee}{k^+} + \log(1-a)$:

$$\text{(L2)} = \frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\mathrm{LO}[\mathcal{B}] + \frac{g^2N_c}{4\pi^2}\log\frac{\vee}{k^+}\mathrm{LO}[\mathcal{B}] + \frac{g^2N_c}{4\pi^2}\log(1-a)\mathrm{LO}[\mathcal{B}].\tag{4.2}$$

The first two terms are the evolution terms of this row (JIMWLK and BFKL lists).

Step 3: the $\log(1-a)$ completes the plus distribution. By Eq. (2.4), $2N_c\,G(1)\log(1-a)$ is the endpoint term of the standard plus distribution, and $2N_cG(1) = 2N_c\mathrm{LO}[\mathcal{B}]$. So

$$\text{(L1)} + \frac{g^2N_c}{4\pi^2}\log(1-a)\mathrm{LO}[\mathcal{B}] = \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,P^{\mathrm{r}}(\xi)\,\mathrm{LO}_\xi[\mathcal{B}], \qquad P^{\mathrm{r}} \equiv \mathcal{P}_{gg} - \frac{\beta_0}{2}\delta(1-\xi)\tag{4.3}$$

(standard plus distribution, Eq. (2.4); $\frac{g^2N_c}{4\pi^2} = \frac{g^2}{8\pi^2}2N_c$).

Step 4: add and subtract $\frac{\beta_0}{2}\delta(1-\xi)$. At $\xi=1$, $\mathrm{LO}_\xi[f]=\mathrm{LO}[f]$:

$$(4.3) = \underbrace{\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{(A)}} + \underbrace{\frac{g^2\beta_0}{16\pi^2}\mathrm{LO}\Big[\frac{2}{\epsilon}+\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{(R)}}.\tag{4.4}$$

Step 5: DGLAP. The hadron is $\frac{d^3N_h}{d^3k} = \int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}\big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\big)$. The $\overline{MS}$ fragmentation function is $D^{\mathrm{bare}}(\zeta) = D(\zeta,\mu_F) + \frac{g^2}{8\pi^2}\big[\frac{2}{\epsilon}+\log\frac{\mu^2}{\mu_F^2}\big]\int_\zeta^1\frac{dz}{z}\mathcal{P}_{gg}(z)D\big(\frac{\zeta}{z},\mu_F\big)$.

- Inserted into LO, the counterterm covers $a\le\zeta\le\zeta'\le1$, the same region as the ξ integral of (A) ($\zeta=\zeta'\xi\ge a$). The standard plus distribution, endpoint $\log(1-a)$ included, is the one of the DGLAP kernel.
- So the pole of (A) cancels, and $\log\frac{\mu^2}{\mu_F^2}$ turns $\ell_\Delta(\mu)$ into $\ell_\Delta(\mu_F)$.
- The leftover $\frac{\epsilon}{2}\ell_\Delta$ of $\frac{d_\perp}{2}$ has no pole in front and vanishes.

$$\text{(A)} \;\to\; -\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big].\tag{4.5}$$

Step 6: running coupling. LO is $\propto g^2$. The $\overline{MS}$ coupling counterterm, $g^2 \to g^2(\mu_R)\big[1 - \frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\big]$ ($\frac{2}{\epsilon} = \frac{1}{\epsilon_{D=4-2\epsilon}}$), adds $-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}$LO. Then, at $\mu = \mu_R$ and with $d_\perp \to 2$,

$$(\mathrm{R}) - \frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\mathrm{LO} = \frac{g^2\beta_0}{16\pi^2}\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big], \tag{4.6}$$

and, with $\alpha_s(Q) = \alpha_s(\mu_R)\big[1 - \frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}\big]$,

$$\mathrm{LO}\big|_{\alpha_s(\mu_R)} + (4.6) = \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + O(\alpha_s^3). \tag{4.7}$$

Here y, y' are the points where the produced gluon crosses the target in the amplitude and in the conjugate amplitude. Through $e^{-ik\cdot(y'-y)}$, $|y'-y| \sim 1/|\mathbf{k}|$. The μ_R dependence cancels: $\partial_{\log\mu_R^2}$ of LO is $-\frac{\alpha_s^2\beta_0}{4\pi}$LO, that of (4.6) is $+\frac{\alpha_s^2\beta_0}{4\pi}$LO.

Step 7: the $\log\frac{\vee}{k^+}$ hidden in (4.5). For $\xi \to a \ll 1$ the phase $e^{-ik\cdot(y'-y)/\xi}$ forces $y' \to y$. The $\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$, with $\frac{d\xi}{\xi^2}$, then behaves like $\frac{d\xi}{\xi}$ and gives $\log\frac{\vee}{k^+}$. Its coefficient is the $y' = y$ term, $\mathrm{LO}_\xi^{\mathrm{sub}}$. Add and subtract it:

$$(4.5) = \underbrace{-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\mathrm{sub}}\big[\ell_\Delta(\mu_F)\big]\Big\}}_{\text{DGLAP, no }\log\frac{\vee}{k^+}} - \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\mathrm{sub}}\big[\ell_\Delta(\mu_F)\big]. \tag{4.8}$$

In $\mathrm{LO}_\xi^{\mathrm{sub}}$ only the phase and the weight depend on y'. So:

- the constant and $\log\mu_F^2$ multiply $\int d^2y'\,e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k}) = 0$ ($\mathbf{k} \neq 0$);
- $\int d^2y'\,e^{-ik\cdot(y'-y)/\xi}\log(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}$ (from $\nabla^2\log r^2 = 4\pi\delta^2(r)$; checked numerically).

Hence $\mathrm{LO}_\xi^{\mathrm{sub}}[\ell_\Delta(\mu_F)] = \mathrm{LO}_\xi^{\mathrm{sub}}[\log(y'-y)^2] = -\frac{4\pi\xi^2}{\mathbf{k}^2}\mathrm{LO}^y$, and

$$-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\mathrm{LO}^y = \frac{g^2N_c}{\pi\mathbf{k}^2}\mathrm{LO}^y\int_a^1 d\xi\frac{1-\xi}{\xi} = \Big[\log\frac{\vee}{k^+} - \Big(1 - \frac{k^+}{\vee}\Big)\Big]H_4. \tag{4.9}$$

- $\log\frac{\vee}{k^+}H_4$ is a BFKL term: it is the $+\log\frac{\vee}{k^+}H_4$ of your BFKL list.
- $-(1-\frac{k^+}{\vee})H_4$ is finite.
- The subtraction term inside DGLAP equals $-\big[\log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\big]H_4$, with no pole and no μ.

Step 8: (L3). $\int_a^{\xi_\Lambda} = \int_a^1 + O(\frac{\Lambda}{k^+})$: the integrand $\propto 1-\xi$ vanishes at $\xi = 1$. At $\xi \to a$, $\mathrm{LO}_\xi^{\mathrm{lp}} = O(\xi^2)$ (the δ^{ij}-traceless part of the kernel has no $\delta^2(\mathbf{k})$ term), so there is no $\log\frac{\vee}{k^+}$. No pole multiplies it, so $d_\perp \to 2$:

$$(\mathrm{L3}) = \frac{g^2N_c}{4\pi^2}\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{\mathrm{lp}}\Big|_{d_\perp=2}. \tag{4.10}$$

Step 9: collect. Steps 2–8 give Eq. (1.1):

- JIMWLK: $\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\mathrm{LO}[\mathcal{B}]$;
- BFKL: $\frac{g^2N_c}{4\pi^2}\log\frac{\vee}{k^+}\mathrm{LO}[\mathcal{B}] + \log\frac{\vee}{k^+}H_4$;
- DGLAP (Step 7, first term);
- running coupling (4.6);
- finite: $-(1-\frac{k^+}{\vee})H_4$+(L3).

5 Checks

check (`checks.py` → `checks.json`)	result
$\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta$ (sympy)	exact
$\frac{g^2N_c}{4\pi^2} = \frac{\alpha_s}{2\pi}2N_c$	exact
cut $\xi \le \xi_\Lambda$ = your plus $+G(1)[\log\frac{k^+}{\Lambda} + \log(1-a)]$, $\Lambda/k^+ = 10^{-5}$	2.6×10^{-5} ($O(\Lambda/k^+)$)
$\int_a^1\frac{1-\xi}{\xi}d\xi = \log\frac{1}{a} - (1-a)$	exact
$2\pi\int rJ_0(qr)\log r^2e^{-\delta r^2}dr \to -\frac{4\pi}{q^2}$ ($q = 1.7$; $\delta = 10^{-2}, 3\times10^{-3}, 10^{-3}$)	-4.410, -4.366, $-4.354 \to -4.348$
$\frac{\alpha_s}{2\pi}2N_c\frac{4\pi}{\mathbf{k}^2} = \frac{g^2N_c}{\pi\mathbf{k}^2}$; H_4 normalisation (`g2iv_rc/compact`)	exact
running coupling after the counterterm; μ_R independence	exact